UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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ASSIGNMENT 2

SECJ3553 – ARTIFICIAL INTELLIGENCE

SECTION 02

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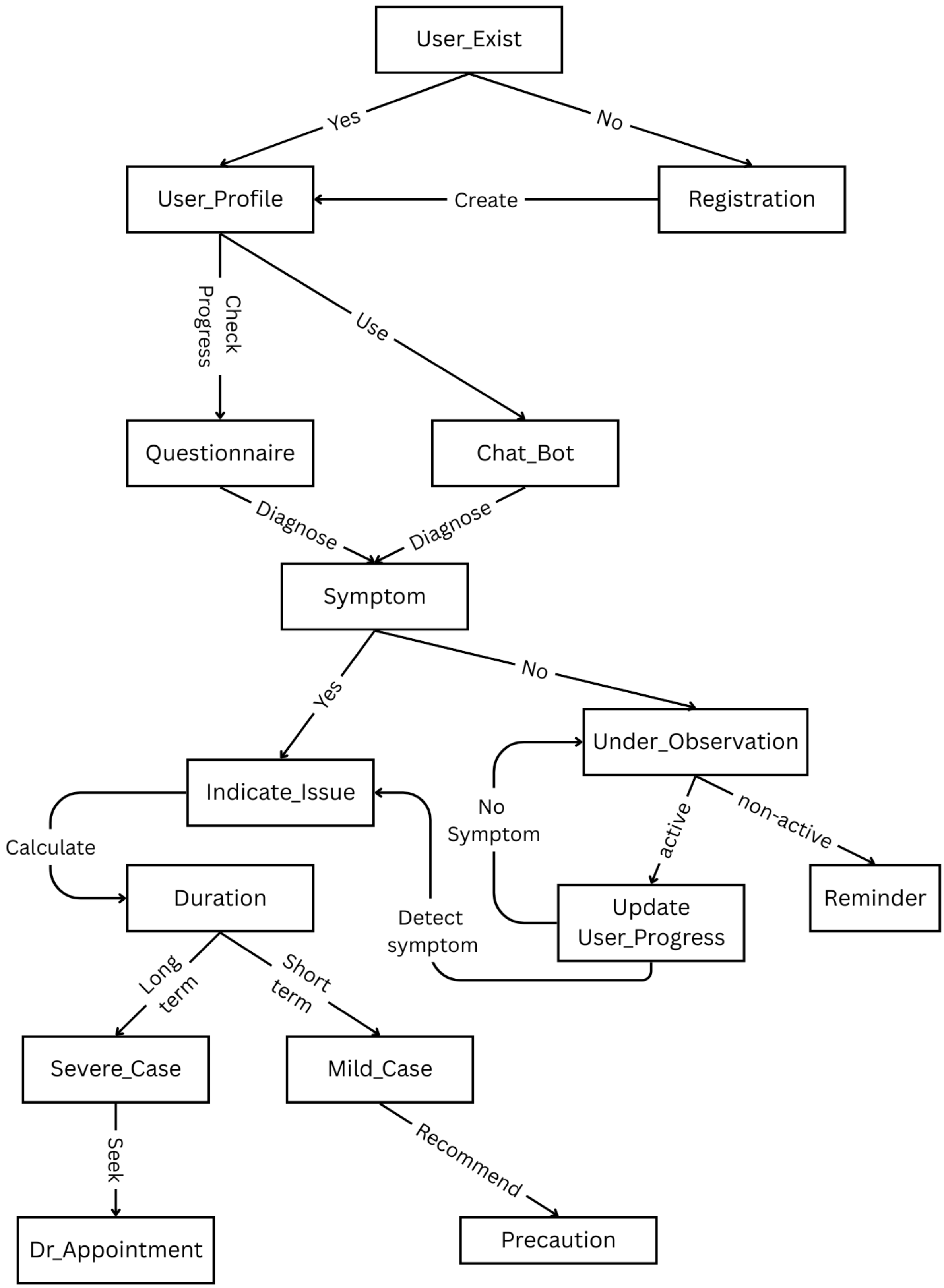

User_Exist
Yes
No
User_Profile
Create
Registration
Check Progress
Use
Questionnaire
Chat_Bot
Diagnose
Diagnose
Symptom
Yes
No
Under_Observation
Indicate_Issue
No Symptom
active
non-active
Calculate
Duration
Update User_Progress
Reminder
Detect symptom
Long term
Short term
Severe_Case
Mild_Case
Seek
Recommend
Dr_Appointment
Precaution

Initial states: User_Exist

Actions: User_Profile, Registration, Questionnaire, Chat_Bot, Symptom, Indicate_Issue, Under_Observation, Duration, Update User_Progress, Severe_Case, Mild_Case

Goals: Dr_Appointment, Precaution, Reminder

Path cost: Yes, No, Create, Check Progress, Use, Diagnose, Active, Non-active, No Symptom, Detect symptom, Calculate, Long term, Short term, Seek, Recommend

Explanation:

The process begins at the Initial State by determining whether a User_Exist. If the user is new, the Registration action will be triggered by the system to Create a User_Profile. The profile of the user will be the central repository for all process and historical progress.

Once the user has activated their profile, the system will move into a diagnostic state where the user can choose to take a Questionnaire or use a Chat_Bot interface. These two interfaces utilize Natural Language Processing (NLP) and Linguistic Patterns to Diagnose users based on their Symptom shown when they are using the interface. From the Symptom obtained (Yes) from the diagnosis, the system will transition to the Indicate_Issue state. In this stage, the systems will Calculate the Duration for evaluation to determine whether it has lasted for long term or short term in order to determine the severity of the case. This transition represents the execution of First-Order-Logic (FOL) as it is used to indicate a specific condition of a disease.

Different severity and duration will lead to different results, where the long term and Severe_Case user will need to Seek for Dr_Appointment. As for users with short term and Mild_Case, the system will Recommend a Precaution for an early precaution. Hence, this search has reached its Goal State.

If no Symptom detected (No), users will enter the Under_Observation state, where this state will utilize Semantic Networks to monitor the engagement of the users. If the user remains active, the system will Update User_Progress, where in this state the system will periodically loop back to Symptom state to update the user's emotional state. Next, if the user is non-active, the system will issue a Reminder to update their emotional state.